

Durval Polónia

Software Engineer

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EXPERIENCE

Junior Software Engineer, Critical Software — Lisbon, Portugal

Feb 2026 – Present

- Maintain and extend the integration API bridging BMW's internal product data with the seller-facing applications used across **200+ markets**, serving roughly **40M requests per day**.
- Develop backend services in **Java** on **Quarkus**, covered by a layered test suite of JUnit unit tests, Cucumber BDD scenarios, and Testcontainers integration tests against real dependencies.
- Redesigned and automated the application infrastructure across multiple **AWS** services as declarative **CloudFormation** templates, replacing legacy import pipelines.
- Extended the **GitHub Actions** CI/CD pipelines.
- Introduced **agentic AI** into day-to-day delivery and prototyped novel agent-assisted development workflows **now used by the team**.

Research Fellow, University of Aveiro — Aveiro, Portugal

Apr 2025 – Jan 2026

- Developed an end-to-end pipeline — capture, labelling, feature extraction, training, evaluation — training system-call-based anomaly detection ML models for serverless environments, reaching **0.97 MCC** at **sub-millisecond average inference**.
- Built an **eBPF** system-call interception layer tracing function behaviour in the Linux kernel at runtime, plus a labelled dataset backed by an in-depth analysis of cloud- and serverless-specific vulnerabilities.
- Authored the resulting detection system, **FADO**, published at **IEEE NOMS 2026**.

Apprentice Backend Developer, Sky Portugal — Aveiro, Portugal

Jul 2024 – Aug 2024

- Developed **Kotlin** backend code for Peacock, Sky's streaming platform, serving **tens of thousands of daily active users**.
- Boosted inter-team cooperation by automating the creation of pull requests for product data changes.
- Documented the full project lifecycle under **Scrum**.

QA & Test Automation Summer Intern, adidas — Porto, Portugal

Jul 2023 – Aug 2023

- Used **AWS** to integrate a new data-analysis module into adidas's existing cloud infrastructure.
- Created data-centric views quantifying the test automation team's delivered value for other internal teams.
- Co-presented the internship to **several hundred** in adidas's Porto tech community.

EDUCATION

MSc, Software Engineering — University of Aveiro, Portugal

Sept 2023 – Dec 2025

Thesis: *Leveraging system call information to provide function-specific protection in serverless environments.*

Erasmus+ Exchange Semester — University of Padova, Italy

Feb 2024 – Jul 2024

BSc, Computer and Informatics Engineering — University of Aveiro, Portugal

Sept 2020 – Jul 2023

PUBLICATIONS

- **FADO: Detecting Anomalies in Serverless Functions via eBPF System Call Tracing.** Durval P. et al. *IEEE Network Operations and Management Symposium (NOMS)*, 2026.

RELEVANT PROJECTS

Personal Finance Platform — Spring Boot microservices

2026

Publicly deployed at big4finance.online | github.com/Durval8/big4

- **Spring Boot 3 / Java 21 + PostgreSQL** core service (transactions, budgets, net-worth analytics) plus a separately deployed **MongoDB**-backed investments service owning its own data.
- Decoupled over **RabbitMQ** with an **outbox pattern**, behind an nginx gateway presenting one origin to a React 18 + TypeScript frontend; unit, web-slice and Testcontainers tests.
- Built through an **agentic-AI workflow centred on context management**: a nine-document versioned specification set (system design, data model, API, testing) maintained as durable context for coding agents, keeping generated code aligned with the intended architecture across sessions.

nvideo — containerised full-stack film catalogue

2026

github.com/Durval8/nvideo

- **Django REST Framework + PostgreSQL** catalogue with automated TMDb ingestion, behind an nginx reverse proxy.
- **Event-driven** reporting: a standalone **Flask + MongoDB** read model kept in sync asynchronously through **Celery** tasks over **RabbitMQ**, isolating analytical load from the transactional path.

SKILLS

Languages: Java, Python, Kotlin, C++, SQL, Bash.

Backend & Testing: Quarkus, Spring Boot, Django / Django REST Framework, Flask, Celery, REST API design, JUnit, Cucumber (BDD), Testcontainers, pytest.

Cloud, Data & DevOps: AWS, Infrastructure as Code, CloudFormation, Docker Compose, CI/CD, Git, GitHub Actions, Linux, nginx, PostgreSQL, MongoDB, Elasticsearch, Kafka, RabbitMQ, event-driven architecture, Agile (Scrum, Kanban).

Agentic AI & ML: Claude Code, MCP, agent skill authoring, subagent design, context management, agent memory, applied ML for anomaly detection.

Languages spoken: Portuguese, English, Spanish — all fluent.